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Department of Electrical Engineering

Class: B.Tech. IIV Sem.

Sub: Generation of Electrical Power (E-224)

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Assignments-3

1. The daily output of a steam power station is 18×10^5 kWh. If the coal consumption is 700 tonnes per day, calculate the thermal efficiency of power station. Assume calorific value of coal 8500 kcal/kg.
2. The overall thermal efficiency of a 40 MW turbo alternator is 30 %. If the load factor of the power station is 50 % and coal burnt has a calorific value of 6800 kCal, determine the following, (i) Coal consumption per kWh (ii) Coal consumption per day
3. In a steam power station the coal consumption is 0.4 kg per kwh output. If the calorific value of the coal is 6800 kcal per kg, boiler efficiency is 70 % and mechanical and electrical efficiency of alternator is 90 %. Prepare roughly the heat balance sheet for the power station.
4. Draw the general layout of thermal power station and discuss merits and de-merits of each.
5. Draw the general scheme of SCADA in thermal power station and explain function of each block.

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